

Week 9

1. file_copy.c

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main(int argc, char *argv[])
{
    int source, destination;
    char buffer[1024];
    ssize_t bytes_read, bytes_written;
    long total_bytes = 0;

    if (argc != 3)
    {
        printf("Usage: %s <source> <destination>\n", argv[0]);
        return 1;
    }

    source = open(argv[1], O_RDONLY);

    if (source < 0)
    {
        perror("Error opening source file");
        return 1;
    }

    destination = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0)
    {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    while ((bytes_read = read(source, buffer, sizeof(buffer))) > 0)
    {
        bytes_written = write(destination, buffer, bytes_read);
```

```
    if (bytes_written != bytes_read)
    {
        perror("Error writing file");
        close(source);
        close(destination);
        return 1;
    }

    total_bytes += bytes_written;
}

if (bytes_read < 0)
{
    perror("Error reading file");
}
else
{
    printf("File copied successfully.\n");
    printf("Bytes copied: %ld\n", total_bytes);
}

close(source);
close(destination);

return 0;
}
```

```
bharath@DESKTOP-6D34C2R: ~$ mkdir -p ~/OSSP/Experiment9
bharath@DESKTOP-6D34C2R: ~$ cd ~/OSSP/Experiment9
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ nano file_copy.c
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ gcc file_copy.c -o file_copy
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ echo "Hello OSSP" > input.txt
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ echo "Linux System Calls using WSL" > input.txt
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ cat input.txt
Hello OSSP
Linux System Calls using WSL
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ ./file_copy input.txt output.txt
File copied successfully.
Bytes copied: 40
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ cat output.txt
Hello OSSP
Linux System Calls using WSL
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ cmp input.txt output.txt
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ diff input.txt output.txt
```

2. lib_copy.c

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int main(int argc, char *argv[])
```

```
{
```

```
    FILE *source;
```

```
    FILE *destination;
```

```
    char buffer[1024];
```

```
    size_t bytes_read;
```

```
    if (argc != 3)
```

```
    {
```

```
        printf("Usage: %s <source> <destination>\n", argv[0]);
```

```
        return 1;
```

```
    }
```

```
    source = fopen(argv[1], "rb");
```

```
    if (source == NULL)
```

```
    {
```

```
        perror("Error opening source file");
```

```
        return 1;
```

```
    }
```

```
    destination = fopen(argv[2], "wb");
```

```

if (destination == NULL)
{
    perror("Error opening destination file");
    fclose(source);
    return 1;
}

while ((bytes_read = fread(buffer, 1, sizeof(buffer), source)) > 0)
{
    if (fwrite(buffer, 1, bytes_read, destination) != bytes_read)
    {
        perror("Error writing file");
        fclose(source);
        fclose(destination);
        return 1;
    }
}

printf("File copied successfully using standard library functions.\n");

fclose(source);
fclose(destination);

return 0;
}

```

```

bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ diff input.txt output.txt
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ nano lib_copy.c
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ gcc lib_copy.c -o lib_copy
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ ./lib_copy input.txt library_output.txt
File copied successfully using standard library functions.

```

3. redirect_io.c

```

#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;

    fd = open("redirected_output.txt",
              O_WRONLY | O_CREAT | O_TRUNC,
              0644);

```

```

if (fd < 0)
{
    perror("Error opening output file");
    return 1;
}

if (dup2(fd, STDOUT_FILENO) < 0)
{
    perror("dup2 failed");
    close(fd);
    return 1;
}

close(fd);

printf("Hello OSSP\n");
printf("Linux System Calls using WSL\n");

return 0;
}

```

```

File copied successfully using standard library functions.
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ nano redirect_io.c
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ gcc redirect_io.c -o redirect_io
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ cat input.txt
Hello OSSP
Linux System Calls using WSL
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ ./redirect_io
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ cat redirected_output.txt
Hello OSSP
Linux System Calls using WSL
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ ls -lh
total 76K
-rwxr-xr-x 1 bharath bharath 16K Oct  1 04:37 file_copy
-rw-r--r-- 1 bharath bharath 1.9K Oct  1 04:37 file_copy.c
-rw-r--r-- 1 bharath bharath 40 Oct  1 04:37 input.txt
-rwxr-xr-x 1 bharath bharath 16K Oct  1 04:41 lib_copy
-rw-r--r-- 1 bharath bharath 1.3K Oct  1 04:41 lib_copy.c
-rw-r--r-- 1 bharath bharath 40 Oct  1 04:41 library_output.txt
-rw-r--r-- 1 bharath bharath 40 Oct  1 04:37 output.txt
-rwxr-xr-x 1 bharath bharath 16K Oct  1 04:42 redirect_io
-rw-r--r-- 1 bharath bharath 1.4K Oct  1 04:42 redirect_io.c
-rw-r--r-- 1 bharath bharath 40 Oct  1 04:42 redirected_output.txt
bharath@DESKTOP-6D34C2R:~/OSSP/Experiment9$ |

```